

Intermediate Macroeconomics

NAME:

Problem Set 3

Question 1: The nice properties of Cobb-Douglas - Part 1

In class we mentioned four properties that we typically require production functions to satisfy. Assume that the production function is $Y = AK^\alpha N^{1-\alpha}$, where Y denotes output, A denotes total factor productivity, K denotes capital, $0 < \alpha < 1$. Show that

1. The marginal products of capital and labor are always positive.
2. The marginal products of capital and labor are diminishing.
3. Complementarity. For example, if the labor input increases, then the marginal product of capital is higher.
4. Constant returns to scale.

Question 2 - Data on labor shares

1. Briefly describe what is the labor share of income
2. Penn World Table (available here <https://www.rug.nl/ggdc/productivity/pwt/?lang=en>) is a data set (in Excel format) that includes a few data series for many countries. In it, the series tagged “labsh” reports the labor share for each country over time. (See the legend tab for definitions of all variables).

Download the data on labor shares for two different countries of your choice and answer the following:

- (a) Plot the series starting 1990 until the most recent data point. Put the year on the horizontal axis and the labor share on the vertical axis.
- (b) For each country, calculate the average labor share from 1990 until the most recent year available.
- (c) Describe whether the labor share (for each country) appears to be stable or appears to involve a distinct trend.
- (d) Now assume that all countries in the world have a Cobb-Douglas production function $Y = K^\alpha N^{1-\alpha}$ where K is capital and N is labor. Assume that α may be different across countries. What is the value for α that you would assign for each country? Show mathematically how α is related to the labor share using the production function.

Question 3 - Total Factor Productivity and Development Accounting

Our production model assumed that there exists a Total Factor Productivity component (TFP, in short), which is exogenous and scales productivity up or down for given levels of capital and labor input. As we mentioned, there are many factors that may affect TFP, which hopefully we will have time to discuss in a few weeks. The purpose of this question is to use real data and attempt to measure TFP from real data.

1. Use the Penn World Table data that you downloaded in the previous question. Pick two countries and use the following series for each country:
 - Real GDP at constant 2017 national prices (in mil 2017US\$), series code rgdpna
 - Capital stock at constant 2017 national prices (in mil 2017US\$), series code rnna
 - TFP at constant national prices (2017=1), series code rtfpna
 - Number of persons engaged (in millions), emp
 - Average annual hours worked by persons engaged, series code avh
 - Human capital index, series code hc
 - Share of labour compensation in GDP at current national prices, series code labsh
2. For each of the two countries calculate the implied TFP from 1990 to the latest period you have, as follows:
 - Calculate the average labor share for each country. Denote this by $1 - \alpha_i$ for each country i .
 - Assume that the production function is Cobb-Douglas: $Y_t = A_t K_t^\alpha N_t^{1-\alpha}$. Use this formula to compute TFP as follows: $A_t = \frac{Y_t}{K_t^\alpha N_t^{1-\alpha}}$, where
 - K_t is the capital stock for country i
 - N_t is the labor input for country i , computed as the product of the number of people engaged and the average hours of work.
 - For each country use the α that you calculated before.
 - Compute the correlation between the resulting TFP series and the one reported in the dataset (series rtfpna). Suppose that the procedure we performed is accurate, do you *expect* the correlation to be high or low? In your data, is it high? low? positive? negative? (don't worry if it does not work according to your expectations... we'll discuss this).
3. (The rest is for discussion in TA session; no need to submit):
 - What happens if you change the weights that you assign to capital and labor such that both countries have the same coefficient?
 - What happens if you change the weights that you assign to capital and labor such that they are allowed to change by country and year using the labor share data?
 - Show mathematically that if you were using the production function above, but the true/actual production function includes one more factor of production, then measuring TFP based on capital and labor alone will be inaccurate.

Question 4 – EXTRA, not for submission – Growth Accounting

Use the data on the same two countries that you chose in the previous question, and calculate TFP growth according to the growth accounting formula:

- Calculate the commutative growth rate of GDP, capital, and labor. For example: $g_Y = \frac{\Delta Y}{Y} = \frac{Y_{2019} - Y_{1990}}{Y_{1990}}$.
- Assume that the elasticity of output with respect to capital is $\epsilon_{Y,K} = \alpha$ and the elasticity of output with respect to labor is $\epsilon_{Y,N} = 1 - \alpha$.
- Calculate the growth rate of TFP according to: $g_A = g_Y - \epsilon_{Y,K} \times g_K - \epsilon_{Y,N} \times g_N$

Compare this growth rate to the growth rate of TFP that results from the development accounting exercise from the previous question.